

Math 242, S19
Exam 2

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	20	
2	20	
3	24	
4	18	
5	18	
Total	100	

Good Luck !

Problem 1. (20 points) Given the following homogeneous second-order differential equation:

$$y'' + 6y' + 13y = 0.$$

(a) Find the general solution of the given differential equation.

Solution: Characteristic equation

$$r^2 + 6r + 13 = 0$$

$$r = \frac{-6 \pm \sqrt{6^2 - 4 \cdot 13}}{2}$$

$$= \frac{-6 \pm \sqrt{16}}{2} = -3 \pm 2j$$

$$\Rightarrow y(x) = e^{-3x}(C_1 \cos 2x + C_2 \sin 2x)$$

(b) Find a particular solution of the equation that satisfies the initial conditions

$$y(0) = 3, \quad y'(0) = 1.$$

Solution: $y'(x) = (-3)e^{-3x}(C_1 \cos 2x + C_2 \sin 2x)$

$$+ e^{-3x}(-2C_1 \sin 2x + 2C_2 \cos 2x)$$

$$= e^{-3x}[C_1(-3 \cos 2x - 2 \sin 2x) + C_2(-3 \sin 2x + 2 \cos 2x)]$$

$$y(0) = 3 \Rightarrow C_1 = 3$$

$$y'(0) = 1 \Rightarrow -3C_1 + 2C_2 = 1 \quad \rightarrow \begin{cases} C_1 = 3 \\ C_2 = 5 \end{cases}$$

$$\Rightarrow y(x) = e^{-3x}(3 \cos 2x + 5 \sin 2x)$$

Problem 2. (20 points) Given a nonhomogeneous linear third-order differential equation:

$$y^{(3)} + 4y'' + 4y' = 6x - 1.$$

(a) Find the complementary solution y_c of the above nonhomogeneous equation (i.e., the general solution of the associated homogeneous equation).

Solution: Characteristic equation

$$r^3 + 4r^2 + 4r = 0$$

$$r(r^2 + 4r + 4) = 0$$

$$r(r+2)^2 = 0$$

$$r_1 = 0, \quad r_2 = r_3 = -2$$

$$\Rightarrow y_c(x) = C_1 + (C_2 + C_3 x)e^{-2x}$$

(b) Then find a particular solution y_p of the nonhomogeneous equation by the method of undetermined coefficient.

Solution: $f(x) = 6x - 1$. thus we have

$$y_p(x) = x^s (Ax + B)$$

$\Rightarrow s=1$ to avoid duplication of terms in $y_c(x)$

$$\Rightarrow y_p(x) = x(Ax + B) = Ax^2 + Bx$$

$$y_p'(x) = 2Ax + B$$

$$y_p''(x) = 2A$$

$$y_p'''(x) = 0$$

$$\left. \begin{array}{l} y_p'(x) = 2Ax + B \\ y_p''(x) = 2A \\ y_p'''(x) = 0 \end{array} \right\} \Rightarrow \begin{array}{l} y_p''' + 4y_p'' + 4y_p' \\ = 0 + 4(2A) + 4(2Ax + B) \\ = 8Ax + (8A + 4B) \equiv 6x - 1 \end{array}$$

$$\Rightarrow \begin{cases} 8A = 6 \\ 8A + 4B = -1 \end{cases} \Rightarrow \begin{array}{l} A = \frac{3}{4} \\ B = -\frac{7}{4} \end{array}$$

$$\text{thus } y_p(x) = \frac{3}{4}x^2 - \frac{7}{4}x$$

Problem 3. (24 points) Find the Laplace or inverse Laplace transforms of the functions:

(a) $\mathcal{L}\{t^3 - 3 \sinh 2t\}$

$$= \frac{3!}{s^4} - 3 \cdot \frac{2}{s^2 - 2^2} = \frac{6}{s^4} - \frac{6}{s^2 - 4}$$

(b) $\mathcal{L}\{e^{-t} + t^3 e^{2t}\}$

$$= \frac{1}{s+1} + \frac{3!}{(s-2)^4} = \frac{1}{s+1} + \frac{6}{(s-2)^4}$$

(c) $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+9)}\right\}$ using the formula $\mathcal{L}^{-1}\left(\frac{F(s)}{s}\right) = \int_0^t f(\tau) d\tau$.

$$F(s) = \frac{1}{s^2+9} \Rightarrow f(t) = \mathcal{L}^{-1}\left\{\frac{1}{s^2+9}\right\} = \frac{1}{3} \sin 3t$$

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{1}{s(s^2+9)}\right\} &= \int_0^t \frac{1}{3} \sin 3\tau d\tau \\ &= -\frac{1}{9} \cos 3\tau \Big|_{\tau=0}^{\tau=t} \end{aligned}$$

$$= -\frac{1}{9} (\cos 3t - 1)$$

(d) $\mathcal{L}^{-1}\left\{\frac{s+8}{(s+3)^2+16}\right\}$

$$\frac{s+8}{(s+3)^2+16} = \frac{s+8}{(s+3)^2+4^2} = \frac{(s+3)+5}{(s+3)^2+4^2}$$

$$\begin{aligned} &= \mathcal{L}^{-1}\left\{\frac{s+8}{(s+3)^2+16}\right\} = \mathcal{L}^{-1}\left\{\frac{s+3}{(s+3)^2+4^2}\right\} + \mathcal{L}^{-1}\left\{\frac{5}{(s+3)^2+4^2}\right\} \\ &= e^{-3t} (\cos 4t + \frac{5}{4} e^{-3t} \sin 4t) \end{aligned}$$

Problem 4. (18 points) Use Laplace transform to solve the initial value problem:

$$x'' + 9x = 0; \quad x(0) = 3, \quad x'(0) = 6.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\begin{aligned}\mathcal{L}\{x''(t)\} &= s^2 X(s) - sX(0) - x'(0) \\ &= s^2 X(s) - 3s - 6\end{aligned}$$

thus

$$\mathcal{L}\{x'' + 9x\} = \mathcal{L}\{0\}$$

$$\mathcal{L}\{x''\} + 9\mathcal{L}\{x\} = 0$$

$$[s^2 X(s) - 3s - 6] + 9X(s) = 0$$

$$(s^2 + 9)X(s) = 3s + 6$$

$$X(s) = \frac{3s + 6}{s^2 + 9}$$

$$\Rightarrow x(t) = \mathcal{L}^{-1}\left\{\frac{3s + 6}{s^2 + 9}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{3s}{s^2 + 3^2}\right\} + \mathcal{L}^{-1}\left\{\frac{6}{s^2 + 3^2}\right\}$$

$$= 3 \cos 3t + 2 \sin 3t$$

Problem 5. (18 points) Use Laplace transform to solve the initial value problem

$$x'' + 2x' - 8x = 2; \quad x(0) = x'(0) = 0.$$

Solution: let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\mathcal{L}\{x'(t)\} = sX(s) - x(0) = sX(s)$$

$$\mathcal{L}\{x''(t)\} = s^2X(s) - sx(0) - x'(0) = s^2X(s)$$

then

$$\mathcal{L}\{x'' + 2x' - 8x\} = \mathcal{L}\{2\}$$

$$\mathcal{L}\{x''\} + 2\mathcal{L}\{x'\} - 8\mathcal{L}\{x\} = \frac{2}{s}$$

$$s^2X(s) + 2sX(s) - 8X(s) = \frac{2}{s}$$

$$(s^2 + 2s - 8)X(s) = \frac{2}{s}$$

$$\Rightarrow X(s) = \frac{2}{s(s^2 + 2s - 8)}$$

$$\begin{aligned} \frac{2}{s(s^2 + 2s - 8)} &= \frac{2}{s(s-2)(s+4)} = \frac{A}{s} + \frac{B}{s-2} + \frac{C}{s+4} \\ &= \frac{A(s-2)(s+4) + Bs(s+4) + Cs(s-2)}{s(s-2)(s+4)} \end{aligned}$$

$$\text{let } s=0, \text{ we get } 2 = A(-2)(4) \Rightarrow A = -\frac{1}{4}$$

$$s=2, \text{ we get } 2 = B(2)(6) \Rightarrow B = \frac{1}{6}$$

$$s=-4, \text{ we get } 2 = C(-4)(-6) \Rightarrow C = \frac{1}{12}$$

Thus

$$\begin{aligned} x(t) &= \mathcal{L}^{-1}\left\{\frac{2}{s(s^2 + 2s - 8)}\right\} \\ &= \mathcal{L}^{-1}\left\{-\frac{1}{4} \frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{6} \frac{1}{s-2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{12} \frac{1}{s+4}\right\} \\ &= -\frac{1}{4} + \frac{1}{6}e^{2t} + \frac{1}{12}e^{-4t} \end{aligned}$$